

# Charge-ensemble first-order shell transition in the pinned P1 functional

TECT verification-first repository  
claim A2-FULL-PRODUCTION-WELLPOSED

first issued 2026-08-03 · this version issued 2026-08-03 · v1.0

## 1. Purpose and scope: result and boundary

R-157 proved that the hash-pinned P1 functional has no nonzero equilibrium in its unconstrained field space. That result did not decide a fixed-charge sector or a chemical-potential deformation. This note solves those two different variational problems exactly. Put

$$Q(\Psi) = \frac{1}{2} \|\Psi\|_{L^2}^2, \quad \Omega_\mu(\Psi) = \mathcal{F}_{\text{P1}}(\Psi) - \mu Q(\Psi). \quad (1.1)$$

On the fixed cubic torus of side  $L = 16$ , define  $\lambda_0$  to be the exact spectral bottom of the P1 quadratic operator and set

$$\rho_* = \frac{43}{216}, \quad c_* = \frac{1849}{86400}, \quad \mu_t = \lambda_0 - c_*. \quad (1.2)$$

Then the exact identity

$$\boxed{\mathcal{F}_{\text{P1}}(\Psi) = \mu_t Q(\Psi) + \frac{1}{2} \langle \Psi, (\mathcal{L} - \lambda_0) \Psi \rangle + \mathcal{F}_{\text{II}}(\Psi) + \frac{\gamma}{6} \int \rho(\rho - \rho_*)^2, dx} \quad (1.3)$$

holds, where  $\rho = \Psi^\dagger \Psi$ . Every term after  $\mu_t Q$  is nonnegative.

Consequently a finite-wave-number plane wave is an exact constrained global minimizer at  $Q = Q_* := 11008/27$ . In the grand-canonical problem, zero is the unique global minimizer below  $\mu_t$ , zero coexists with nonzero shell minimizers at  $\mu_t$ , and a nonzero global minimizer wins above  $\mu_t$ . The charge jumps from zero to  $Q_*$  before the zero field loses linear stability, so this is an exact first-order transition.

This result is deliberately not called a physical-vacuum theorem. Neither a conserved physical charge nor an external reservoir producing  $\mu$  has been derived. At the coexistence point

$$\mathcal{F}_{\text{P1}}(\Psi_*) = \mu_t Q_* > 0 = \mathcal{F}_{\text{P1}}(0), \quad (1.4)$$

so the state does not beat the neutral zero reference in the original functional. Its gauge-invariant density and all registered internal bilinears are uniform. The finite wave number is a common-phase winding, not a proof of BCC, a density crystal, or any unique morphology.

## 2. Exact finite-torus quadratic bottom

Write the quadratic operator as

$$\mathcal{L} = (r + Z(-\Delta) + Y(-\Delta)^2)I + M, \quad M = \begin{pmatrix} 1/10 & -1/20 & -1/20 \\ -1/20 & 13/100 & -1/20 \\ -1/20 & -1/20 & 17/100 \end{pmatrix}. \quad (2.1)$$

The characteristic polynomial of  $M$  is

$$p(t) = t^3 - \frac{2}{5}t^2 + \frac{223}{5000}t - \frac{3}{3125}. \quad (2.2)$$

An exact Sturm certificate isolates its smallest root  $m_*$  in

$$\frac{2811617233511}{10^{14}} < m_* < \frac{2811617233512}{10^{14}}. \quad (2.3)$$

For a Fourier index  $n \in \mathbb{Z}^3$ ,

$$|k_n|^2 = \frac{\pi^2}{64}|n|^2, \quad D(x) = r + Zx + Yx^2.$$

The exact rational enclosure of  $\pi$  in the executable certificate proves

$$\frac{5}{2} < \frac{-Z/(2Y)}{\pi^2/64} < \frac{7}{2}. \quad (2.4)$$

Since  $|n|^2$  is a nonnegative integer and  $3 = 1^2 + 1^2 + 1^2$ , the unique radial minimum on the torus is  $|n|^2 = 3$ . It contains the eight vectors

$$n = (\pm 1, \pm 1, \pm 1). \quad (2.5)$$

Thus

$$\lambda_0 = D(3\pi^2/64) + m_*, \quad (2.6)$$

with the certified enclosure

$$0.28811617234458494031 < \lambda_0 < 0.28811617234459494032. \quad (2.7)$$

This is a finite-volume commensurability statement. It neither selects a body-centred cubic density pattern nor removes the continuous shell degeneracy in an infinite-volume limit.

### 3. Completion and saturation

The P1 local potential is

$$f(\rho) = \frac{\lambda}{4}\rho^2 + \frac{\gamma}{6}\rho^3, \quad \lambda = -\frac{43}{100}, \quad \gamma = \frac{81}{50}. \quad (3.1)$$

Direct coefficient matching gives

$$\frac{\lambda_0}{2}\rho + f(\rho) = \frac{\mu_t}{2}\rho + \frac{\gamma}{6}\rho(\rho - \rho_*)^2, \quad \rho_* = -\frac{3\lambda}{4\gamma}, \quad c_* = \frac{3\lambda^2}{16\gamma}. \quad (3.2)$$

Adding the nonnegative spectral remainder and the positive-semidefinite Class-II energy proves (1.3).

Let  $u_*$  be a unit eigenvector of  $M$  for  $m_*$  and take any one of the eight indices (2.5). The field

$$\Psi_*(x) = \sqrt{\rho_*} u_* e^{i(2\pi/L)n \cdot x} \quad (3.3)$$

lies in the quadratic ground eigenspace. Moreover  $\rho$  is constant and every  $m_A = \Psi^\dagger T_A \Psi$  is constant. Hence

$$J_A = \nabla m_A = 0, \quad K_A = 0, \quad \mathcal{F}_{\text{II}}(\Psi_*) = 0. \quad (3.4)$$

All remainders in (1.3) vanish. Since the torus volume is  $16^3$ ,

$$Q_* = \frac{16^3 \rho_*}{2} = \frac{11008}{27}. \quad (3.5)$$

The executable enclosure gives

$$0.26671570938162197734 < \mu_t < 0.26671570938163197735,$$

and

$$108.74098255084795 < \mathcal{F}_{\text{P1}}(\Psi_*) < 108.74098255085204. \quad (3.6)$$

In particular, the original neutral-reference energy difference is strictly positive.

## 4. Fixed-charge theorem

For any prescribed  $Q$ , put  $\bar{\rho} = 2Q/16^3$ . The exact Bregman identity is

$$f(\rho) - f(\bar{\rho}) - f'(\bar{\rho})(\rho - \bar{\rho}) = \frac{(\rho - \bar{\rho})^2}{12}(4\gamma\bar{\rho} + 2\gamma\rho + 3\lambda). \quad (4.1)$$

If  $\bar{\rho} \geq \rho_*$ , the bracket is nonnegative. On the affine constraint  $\int \rho = 2Q$ , the linear Bregman term integrates to zero. Therefore the difference from a ground-shell plane wave of constant density  $\bar{\rho}$  is the sum of the nonnegative spectral remainder,  $\mathcal{F}_{\text{II}}$ , and the nonnegative integral (4.1). Such a plane wave is an exact constrained global minimizer for every

$$\frac{2Q}{16^3} \geq \rho_*. \quad (4.2)$$

At equality, this proves the  $Q_*$  assertion. The form of the constrained global minimizer below  $Q_*$  is not decided here.

The weak Euler–Lagrange equation for a constrained critical point is

$$D\mathcal{F}_{\text{P1}}(\Psi)[v] = \mu \langle \Psi, v \rangle_{\mathbb{R}} \quad \text{for all } v \in H^2. \quad (4.3)$$

Equivalently,

$$Y\Delta^2\Psi - Z\Delta\Psi + (r\text{I} + M)\Psi + \lambda\rho\Psi + \gamma\rho^2\Psi + R_{\text{II}}(\Psi) = \mu\Psi. \quad (4.4)$$

For (3.3),  $R_{\text{II}} = 0$  and  $\mu = \mu_t$ . Thus R-157 is not contradicted:  $D\mathcal{F}_{\text{P1}}(\Psi_*) = \mu_t\Psi_* \neq 0$ , so this is not an unconstrained critical point.

## 5. Grand-canonical transition and exact ordering

Subtracting  $\mu Q$  from (1.3) gives

$$\Omega_\mu(\Psi) = (\mu_t - \mu)Q + \frac{1}{2}\langle \Psi, (\mathcal{L} - \lambda_0)\Psi \rangle + \mathcal{F}_{\text{II}}(\Psi) + \frac{\gamma}{6} \int \rho(\rho - \rho_*)^2. \quad (5.1)$$

Hence:

- if  $\mu < \mu_t$ , zero is the unique global minimizer;
- if  $\mu = \mu_t$ , zero and nonzero ground-shell states of density  $\rho_*$  coexist;
- if  $\mu > \mu_t$ , (3.3) has negative grand potential, while sextic coercivity supplies a global minimizer, which must be nonzero.

The zero-field Hessian is  $\mathcal{L} - \mu\text{I}$ , so its linear spinodal is

$$\mu_{\text{lin}} = \lambda_0. \quad (5.2)$$

On the single ground-plane-wave amplitude family, stationary densities obey

$$\gamma\rho^2 + \lambda\rho + (\lambda_0 - \mu) = 0. \quad (5.3)$$

The amplitude branch appears at

$$\mu_{\text{sn}} = \lambda_0 - \frac{\lambda^2}{4\gamma} = \lambda_0 - \frac{1849}{64800}, \quad (5.4)$$

whereas coexistence occurs at

$$\mu_t = \lambda_0 - \frac{1849}{86400}. \quad (5.5)$$

Thus

$$\boxed{\mu_{\text{sn}} < \mu_t < \mu_{\text{lin}}}. \quad (5.6)$$

At coexistence the two stationary densities are

$$\rho_+ = \frac{43}{216}, \quad \rho_- = \frac{43}{648}. \quad (5.7)$$

The global charge changes discontinuously from zero to  $Q_*$  while zero is still a strict local minimum. This closes the first-order verdict. Equation (5.4) is only an amplitude-family saddle-node statement; this note does not claim full-space stability for every off-coexistence plane-wave branch.

## 6. Devil’s-advocate audit and physical interpretation

1. **Objection: fixed charge proves a spontaneous TECT vacuum. UPHELD against that inference.** The charge sector removes zero from the comparison class. No microscopic conservation law, superselection rule, or cosmological initial condition selecting  $Q \neq 0$  has been derived.
2. **Objection: the chemical potential is already a parameter of P1. UPHELD as false.** The term  $-\mu Q$  changes the variational problem. A reservoir or a derived effective ensemble is a new hypothesis.
3. **Objection: a nonzero torus wave vector establishes a spatial crystal. UPHELD as false.** For (3.3),  $\rho$  and every  $m_A$  are constant. The theorem selects a common-phase winding in this representation, not a gauge-invariant density modulation, BCC star, or unique lattice.
4. **Objection: local gauge completion may remove the phase wave. UPHELD as an open physical gate.** P1 has a global phase symmetry but no dynamical gauge field. A local gauge model could gauge away or screen this configuration and must be analysed as a distinct candidate.
5. **Objection: the canonical A2 gradient flow realizes the fixed-charge transition. UPHELD as false.** That dissipative flow does not generally conserve  $Q$ . Projected relaxation or Hamiltonian dynamics would require separate well-posedness and stability theorems.
6. **Objection: the Class-II cross term spoils saturation. DISMISSED for the states (3.3).** Both registered currents vanish pointwise, while the complete coefficient matrix is positive semidefinite.
7. **Objection: the body-diagonal shell is regulator independent. UPHELD as false.** The integer shell follows from the side-16 torus. Robustness under  $L$ , cutoff, and infinite-volume limits is not proved.
8. **Objection: exact algebra can hide a sign, factor, or convention error. VALID WITH MITIGATION.** A SymPy/Sturm implementation and an independent standard-library rational implementation reconstruct all inputs separately. External review is invited for the real-pairing factor  $1/2$ , Fourier normalization, charge normalization, matrix eigenvalue enclosure, quartic–sextic signs, Bregman factor, units, finite-volume dependence, hardcode masking, and the  $Q \rightarrow 0$  and  $\mu \rightarrow -\infty$  limits.

The truth-first candidate verdict is therefore two-sided. The P1 polynomial and torus spectrum do contain a mathematically exact finite-shell first-order mechanism once a charge ensemble is imposed. The same calculation also shows why it is not yet the sought physical explanation: the ordered state is above the neutral reference, its ensemble provenance is absent, and its registered local observables are uniform.

## 7. Reproduction and result footer

Run from the repository root:

```
$py = "E:\Dev\TECT.venv\Scripts\python.exe"  
& $py codes/foundations/a2_charge_ensemble_first_order_shell_transition.py  
& $py codes/foundations/a2_charge_ensemble_first_order_shell_transition_independent.py  
& $py codes/foundations/a2_charge_ensemble_first_order_shell_transition_verify.py
```

The primary certificate uses exact SymPy polynomial and interval algebra and must report 35/35 PASS. The independent certificate uses only Python's standard-library rational arithmetic and must report 24/24 PASS. The integrated verifier reruns both into temporary files, compares immutable artifacts and exact constants, retains the earlier A2/R-157 regression, and audits the note, PDF, claim, task, exploration, changelog, theorem-map, and generated-ledger surfaces.

```
Result ID: A2-CHARGE-ENSEMBLE-FIRST-ORDER-SHELL-TRANSITION  
Ledger ID: R-158  
Claim: A2-FULL-PRODUCTION-WELLPOSED.  
Precise statement: for the hash-pinned P1 functional on the side-16 torus,  
    after defining  $Q = ||\Psi||_{L^2}^2/2$ , the exact decomposition (1.3) holds.  
    At  $Q = 11008/27$  a finite-shell plane wave is a constrained global minimizer.  
    For  $\Omega_\mu = F_{P1} - \mu Q$  there is exact first-order coexistence at  
     $\mu_t = \lambda_{0-1849}/86400$ , before the zero-field spinodal  $\lambda_{0-}$ .  
Scope: Hash-pinned P1/A2 static functional, side-16 torus, H2 fields,  
    fixed  $Q = ||\Psi||_{L^2}^2/2$  or  $\Omega_\mu = F_{P1} - \mu Q$ , and  $\eta_{\text{shell}} = 0$ .  
Dependencies: A1-PRODUCTION-FUNCTIONAL-REALISATION, the named  
    A2-H3-CANONICAL-PRODUCTION-FUNCTIONAL hypothesis, A2 continuum theorem,  
    and R-157 as the unconstrained boundary.  
Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED, CONDITIONAL.  
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe  
    codes/foundations/a2_charge_ensemble_first_order_shell_transition_verify.py  
Expected output: Integrated verifier PASS; primary 35/35; independent 24/24;  
    R-157 and legacy A2 regressions PASS.  
Falsification gate: Any exact internal/shell/decomposition/Bregman/Class-II,  
    child-parity, artifact-hash, record, PDF, R-157, or legacy-A2 gate fails.  
Tier before / after: T6 / T6; no promotion.  
No-overclaim statement: The charge or chemical potential is imposed, not  
    physically derived;  $F_{P1}[\Psi] > F_{P1}[0]$ , registered local observables are  
    uniform, and no physical vacuum, BCC, unique morphology, infinite-volume,  
    A7, quantum, T-050/A13, or Sector-A conclusion follows.  
Next required action: Freeze PA-M1-Q as conditional; derive charge/reservoir  
    provenance and a gauge-invariant observable, and compare PA-M5-NL3-SV in  
    T-054 before survivor-only T-053 work.  
Result footer.
```